

# Challenge Write-Up: sudo make me a sandwich

## Challenge Information

|                     |                         |
|---------------------|-------------------------|
| Student Name        | Keyzia Joy Pascua       |
| Section / Class     | BSIT 3 - Networking     |
| Challenge Name      | sudo make me a sandwich |
| Category            | General Skills          |
| Points / Difficulty | Easy                    |
| Date Solved         | August 11, 2026         |
| Flag Format Given   | picoCTF{...}            |

## 1. Challenge Description

sudo make me a sandwich is a straightforward Linux file-permissions/enumeration challenge. The prompt gives SSH credentials to a remote server and simply asks whether the flag can be read — no exploit is hinted at, so the task is really about logging in, checking what privilege level that account has, and then searching the filesystem for wherever the flag file actually lives, since it is not necessarily in the obvious location.

## 2. Initial Analysis / Reconnaissance

After connecting over SSH with the given credentials, the session drops into Emacs' welcome/tutorial screen rather than a plain shell, which is presented as the account's default “login shell”. Dropping to a real shell from inside Emacs (or simply working from the shell prompt beneath it) and running whoami confirms the connected account is root — an unusually high privilege level for a CTF login, which suggests the flag will simply be sitting somewhere readable rather than requiring privilege escalation.

## 3. Tools and Commands Used

| Tool / Command                                 | What it was used for                                                                                | Result / Output                                        |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| ssh -p 51694 ctf-player@green-hill.picoctf.net | Log into the challenge instance with the password given in the prompt                               | Connected; session opened into an Emacs welcome screen |
| whoami                                         | Check which user account the shell is actually running as                                           | root                                                   |
| cat /root/flag.txt                             | Try the most obvious location for the flag first, given a root shell                                | No such file or directory                              |
| find / -name flag.txt 2>/dev/null              | Search the whole filesystem for a file literally named flag.txt, discarding permission-denied noise | /home/ctf-player/flag.txt                              |
| cat /home/ctf-player/flag.txt                  | Read the flag file at the location find revealed                                                    | Printed the flag                                       |

## 4. Solution Walkthrough

### Step 1

Connect to the instance with the SSH command given on the challenge page: `ssh -p 51694 ctf-player@green-hill.picoctf.net`, using the supplied password. The session lands not in a normal shell prompt but inside GNU Emacs' startup screen, since Emacs has apparently been set as this account's default shell — a shell prompt is still reachable underneath it (via C-x C-c to exit Emacs cleanly, or the `eshell/shell-command` features Emacs provides).

### Step 2

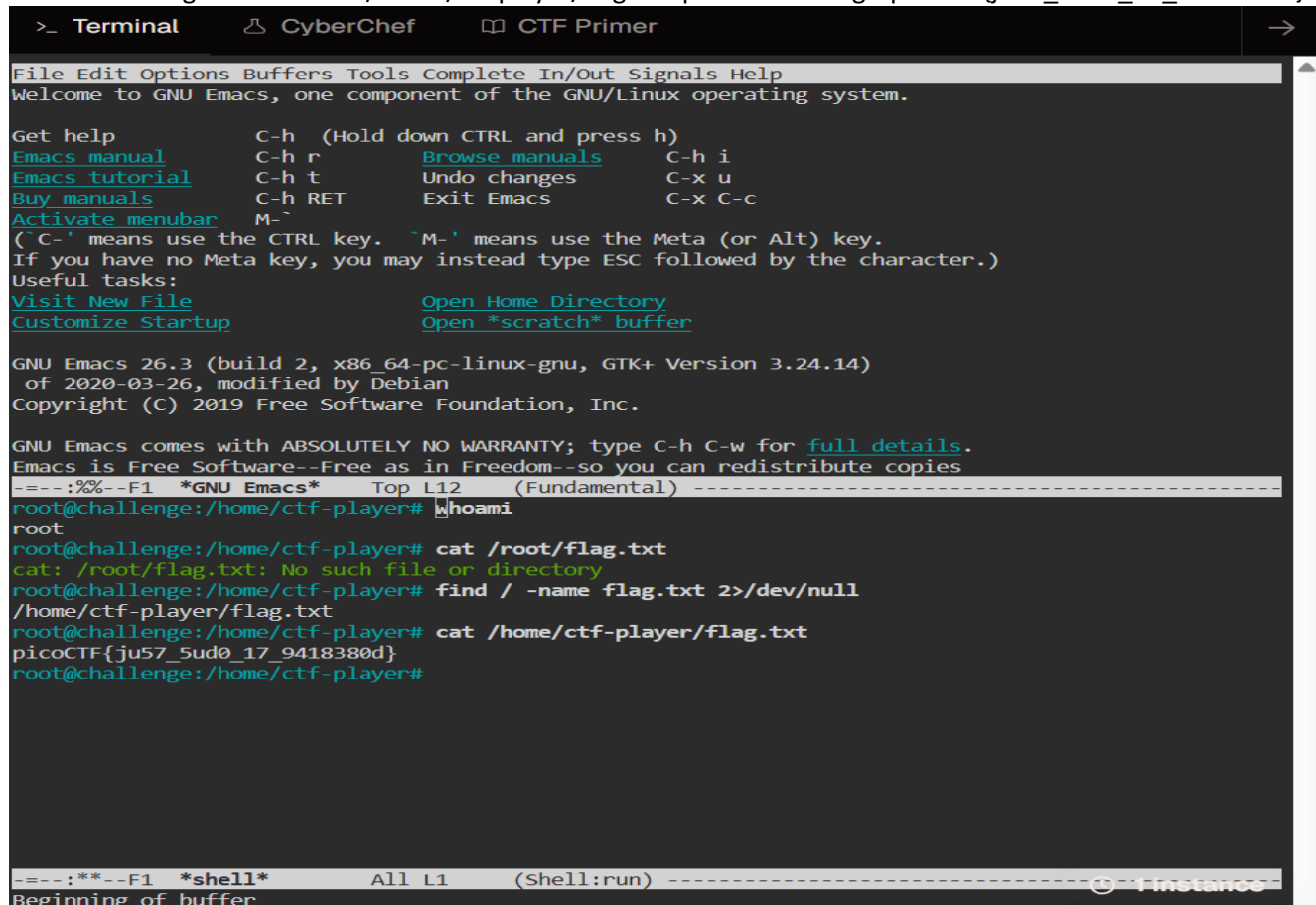
Once at a shell prompt, run `whoami` to see which account is actually connected. The result is `root`, meaning the login is already fully privileged — there is no need to escalate, only to find where the flag file is kept.

### Step 3

Try the obvious guess first: `cat /root/flag.txt`. This fails with “No such file or directory”, so the flag is not sitting directly in `root`'s home directory and needs to be located elsewhere on the filesystem.

### Additional steps (add as needed)

Search the entire filesystem for the file by name: `find / -name flag.txt 2>/dev/null` (redirecting `stderr` to `/dev/null` hides the inevitable “Permission denied” noise from directories the search can't read). This turns up `/home/ctf-player/flag.txt` — the flag was sitting in the login account's own home directory the whole time, not `root`'s. Reading it with `cat /home/ctf-player/flag.txt` prints the flag: `picoCTF{ju57_5ud0_17_9418380d}`.



```
>_ Terminal  CyberChef  CTF Primer  →

File Edit Options Buffers Tools Complete In/Out Signals Help
Welcome to GNU Emacs, one component of the GNU/Linux operating system.

Get help          C-h (Hold down CTRL and press h)
Emacs manual      C-h r      Browse manuals    C-h i
Emacs tutorial    C-h t      Undo changes      C-x u
Buy manuals       C-h RET    Exit Emacs        C-x C-c
Activate menubar  M-`

(`C-' means use the CTRL key.  `M-' means use the Meta (or Alt) key.
If you have no Meta key, you may instead type ESC followed by the character.)
Useful tasks:
Visit New File          Open Home Directory
Customize Startup       Open *scratch* buffer

GNU Emacs 26.3 (build 2, x86_64-linux-gnu, GTK+ Version 3.24.14)
of 2020-03-26, modified by Debian
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GNU Emacs comes with ABSOLUTELY NO WARRANTY; type C-h C-w for full details.
Emacs is Free Software--Free as in Freedom--so you can redistribute copies
---:~::~F1 *GNU Emacs* Top L12 (Fundamental) -----
root@challenge:/home/ctf-player# whoami
root
root@challenge:/home/ctf-player# cat /root/flag.txt
cat: /root/flag.txt: No such file or directory
root@challenge:/home/ctf-player# find / -name flag.txt 2>/dev/null
/home/ctf-player/flag.txt
root@challenge:/home/ctf-player# cat /home/ctf-player/flag.txt
picoCTF{ju57_5ud0_17_9418380d}
root@challenge:/home/ctf-player#

---:~::~F1 *shell* All L1 (Shell:run) ----- Instance
Beginning of buffer
```

*Terminal session showing `whoami`, the `find` search, and reading the flag.*

## 5. Flag

```
picoCTF{ju57_5ud0_17_9418380d}
```

## 6. Key Concept / What I Learned

The key concept is basic Linux filesystem enumeration: when a flag or target file's exact location isn't given, `find / -name <filename> 2>/dev/null` is the fastest way to locate it system-wide without getting buried in permission-denied errors, and it is worth checking `whoami` early since the current privilege level (here, an unexpected root shell) shapes where and how easily a target file can be read. It's also a reminder not to assume a file lives in the “obvious” directory (like the current user's home) just because that user happens to be root.

## 7. References

None beyond the challenge's own prompt and standard documentation for `find`, `ssh`, and GNU Emacs.